

Chapter 9

Angular Momentum

Gravity (2013)

- Science fiction thriller film directed by Alfonso Cuarón
- The story is about American astronauts who attempt to return to Earth after the destruction of their Space Shuttle in orbit.



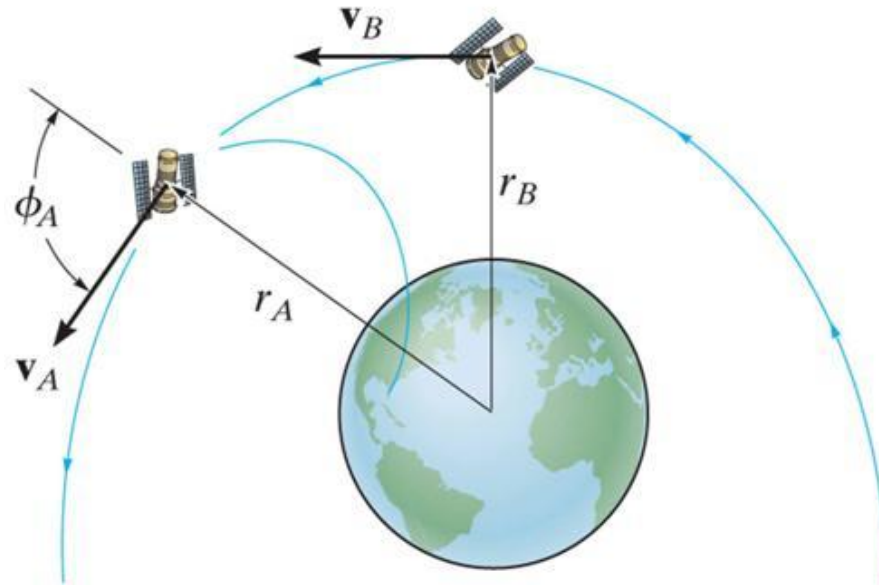
Watch the clips and answer the questions

- How does the accident happen? (e.g., the speed, the mass, momentum)
- Why does the astronaut keep spinning?
- How did the astronaut stop?

Introduction

- In this chapter, we'll introduce some new concepts, such as torque, angular momentum and angular momentum conservation, to deepen our understanding of rotational motion.

Angular momentum: introduction



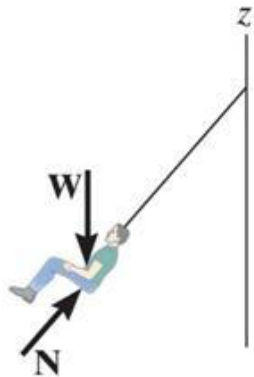
Planets and most satellites move in elliptical orbits. This motion is caused by gravitational attraction forces. Since these forces act in pairs, the sum of the moments of the forces acting on the system will be zero. This means that angular momentum is conserved.

If the angular momentum is constant, does it mean the linear momentum is also constant? Why or why not?

Angular momentum: introduction



The passengers on the amusement park ride experience conservation of angular momentum about the axis of rotation (the z -axis). As shown on the free-body diagram, the line of action of the normal force, N , passes through the z -axis and the weight's line of action is parallel to it. Therefore, the sum of moments of these two forces about the z -axis is zero.

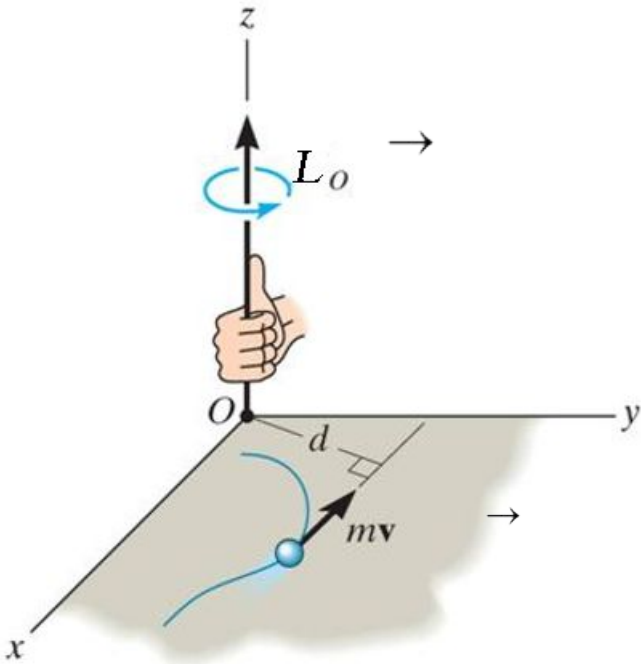


If the passenger moves away from the z -axis, will his speed increase or decrease? Why?

Angular momentum

The angular momentum of a particle about point O is defined as the “moment” of the particle’s linear momentum about O.

(In physics typically denoted by $\vec{L}$, but $\vec{H}$ is also used in some books)



$$\mathbf{L}_O = \mathbf{r} \times m\mathbf{v} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ r_x & r_y & r_z \\ mv_x & mv_y & mv_z \end{vmatrix}$$

The magnitude of $\mathbf{L}_O$ is $(\vec{L}_O)_z = mvd$

Angular momentum

The resultant force acting on the particle is equal to the time rate of change of the particle's linear momentum.

$$\vec{F} = \sum_j \vec{F}_j = \frac{d}{dt} \vec{p} = m \frac{d}{dt} \vec{v}$$

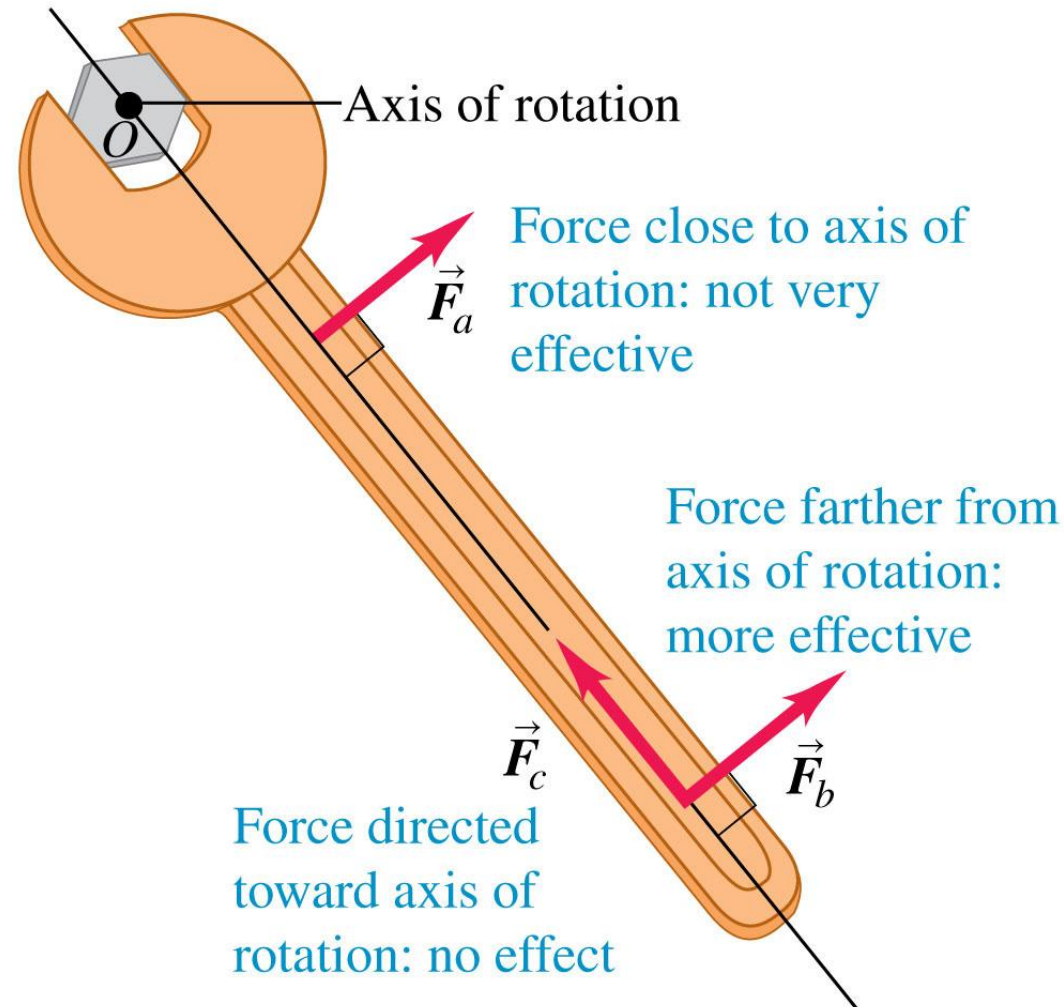
We can prove that the resultant *moment* acting on the particle about point O (also called *torque*, denoted by τ or M) is equal to the time rate of change of the particle's angular momentum about point O or

$$\vec{\tau}_o = \sum_j \vec{r} \times \vec{F}_j = \frac{d}{dt} \vec{L}_o = m \frac{d}{dt} (\vec{r} \times \vec{v})$$

The SI unit of torque is $\text{N}\cdot\text{m} = \text{kg m}^2/\text{s}^2$

Loosen a bolt

- Which of the three equal-magnitude forces in the figure is most likely to loosen the bolt?

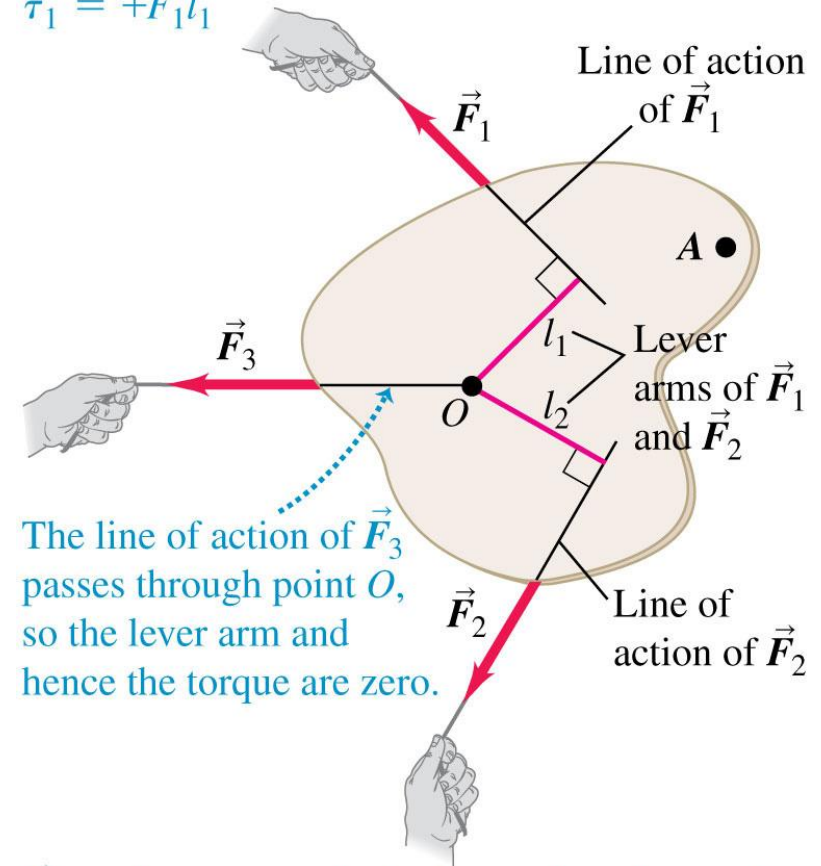


Torque

- The **line of action** of a force is the line along which the force vector lies.
- The **lever arm** for a force is the perpendicular distance from O to the line of action of the force.
- The torque of a force with respect to O is the product of the force and its lever arm.

$\vec{F}_1$ tends to cause *counterclockwise* rotation about point O , so its torque is *positive*:

$$\tau_1 = +F_1 l_1$$

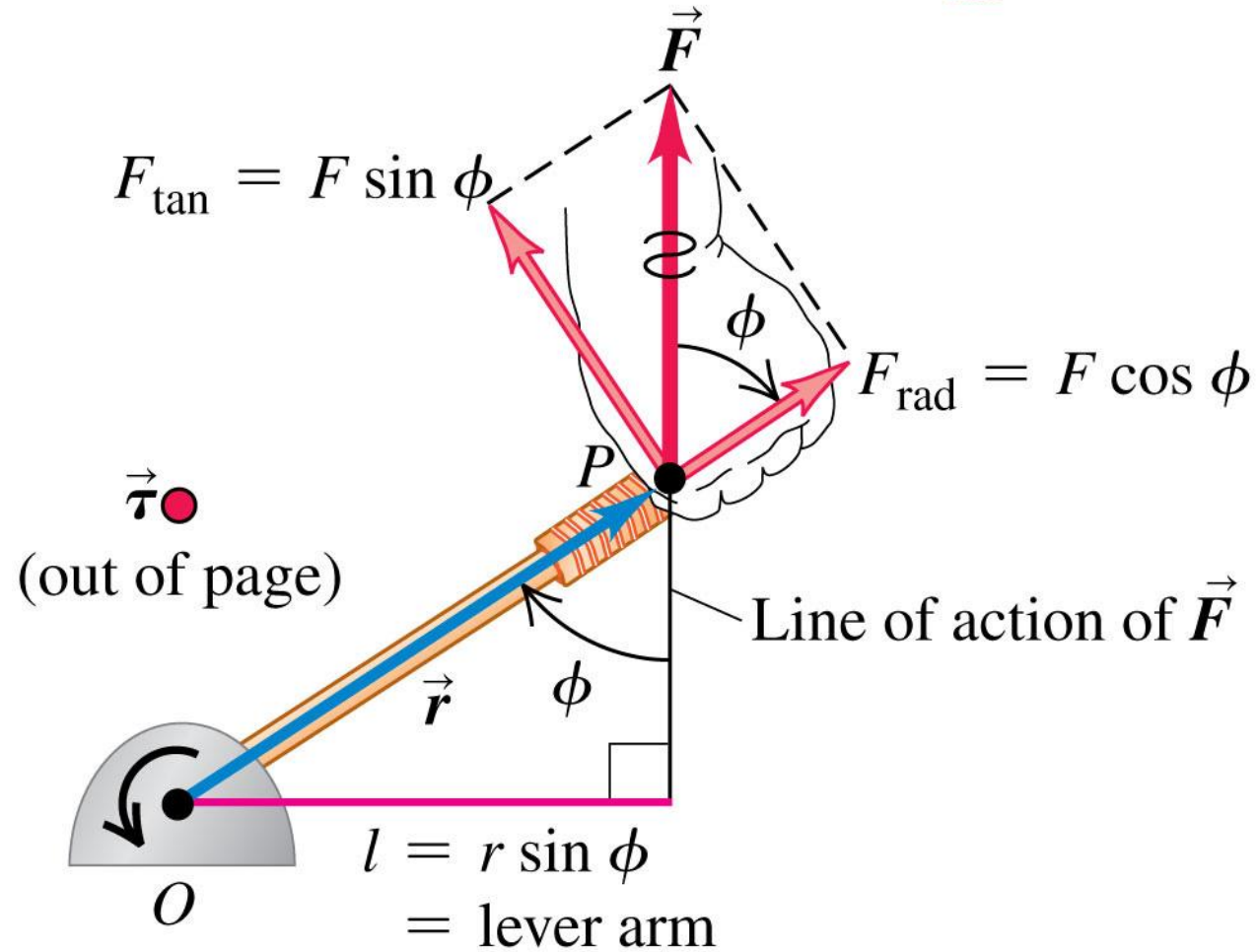


$\vec{F}_2$ tends to cause *clockwise* rotation about point O , so its torque is *negative*: $\tau_2 = -F_2 l_2$

Three ways to calculate torque

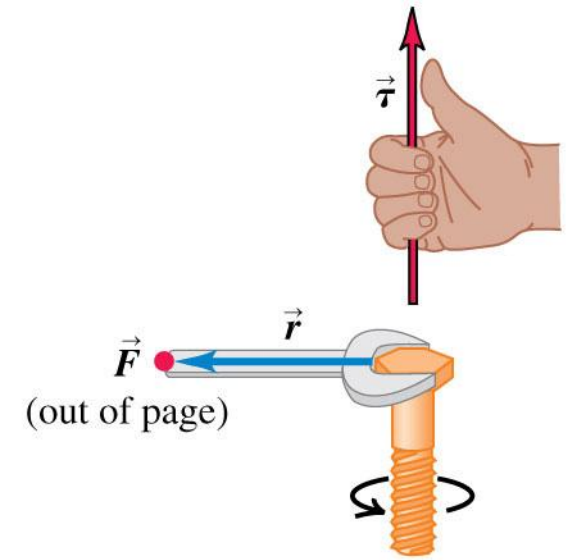
Three ways to calculate torque:

$$\tau = Fl = rF \sin \phi = F_{\text{tan}} r$$



Torque as a vector

- Torque can be expressed as a vector using the vector product.
- If you curl the fingers of your right hand in the direction of the force around the rotation axis, your outstretched thumb points in the direction of the torque vector.



Torque vector due to force $\vec{F}$ relative to point O $\vec{\tau} = \vec{r} \times \vec{F}$ **Vector from O to where $\vec{F}$ acts** **Force $\vec{F}$**

Principle of angular impulse and momentum

Considering the relationship between moment and time rate of change of angular momentum

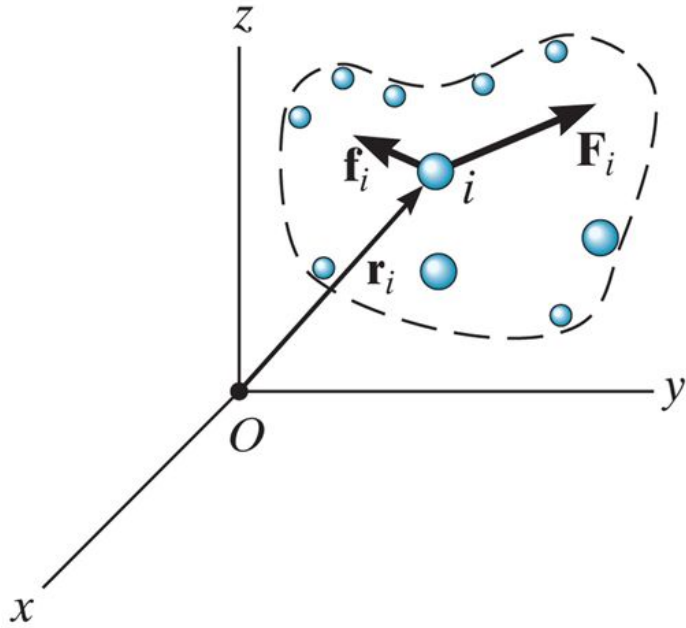
$$\vec{\tau}_o = \frac{d}{dt} \vec{L}_o$$

By integrating between the time interval t_1 to t_2

$$\sum \int_{t_1}^{t_2} \vec{\tau}_o dt = (\vec{L}_o)_2 - (\vec{L}_o)_1$$

This equation is referred to as the **principle of angular impulse and momentum**. The second term on the left side, $\sum \int \tau_o dt$, is called the **angular impulse**. In cases of 2D motion, it can be applied as a scalar equation using components about the z-axis.

Angular momentum of a system of particles



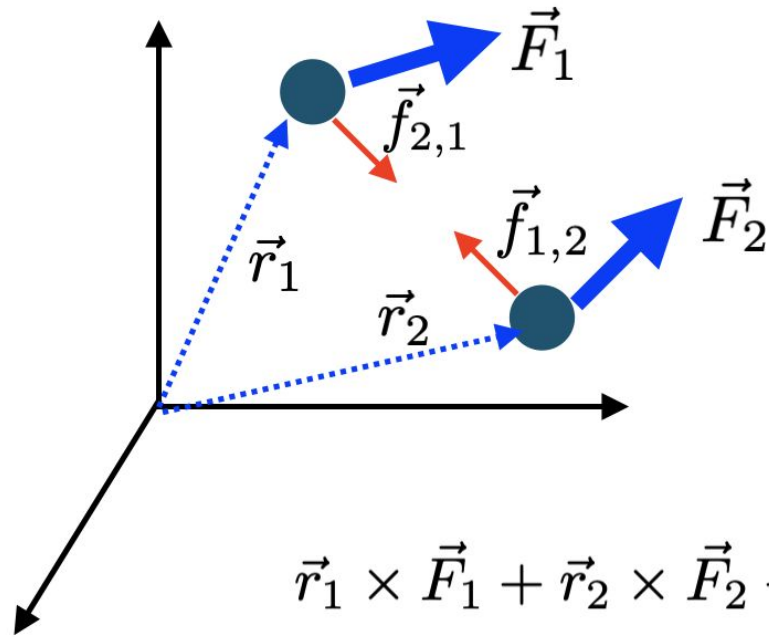
The same form of the equation can be derived for the system of particles.

The forces acting on the i-th particle of the system consist of a resultant external force $\mathbf{F}_i$ and a resultant internal force $\mathbf{f}_i$.

The torques of these forces for the particles are

$$\sum_j \vec{r}_i \times \vec{F}_{j,i} + \sum_{j \neq i} \vec{r}_i \times \vec{f}_{j,i} = \dot{\vec{L}}_{i,o}$$

Angular momentum of a system of particles



Consider a system with two particles:

$$\vec{r}_1 \times \vec{F}_1 + \vec{r}_1 \times \vec{f}_{2,1} = \dot{\vec{L}}_{1,o}$$

$$\vec{r}_2 \times \vec{F}_2 + \vec{r}_2 \times \vec{f}_{1,2} = \dot{\vec{L}}_{2,o}$$

The net angular momentum change

$$\vec{r}_1 \times \vec{F}_1 + \vec{r}_2 \times \vec{F}_2 + \vec{r}_1 \times \vec{f}_{2,1} + \vec{r}_2 \times \vec{f}_{1,2} = \frac{d}{dt} (\vec{L}_{1,o} + \vec{L}_{2,o})$$

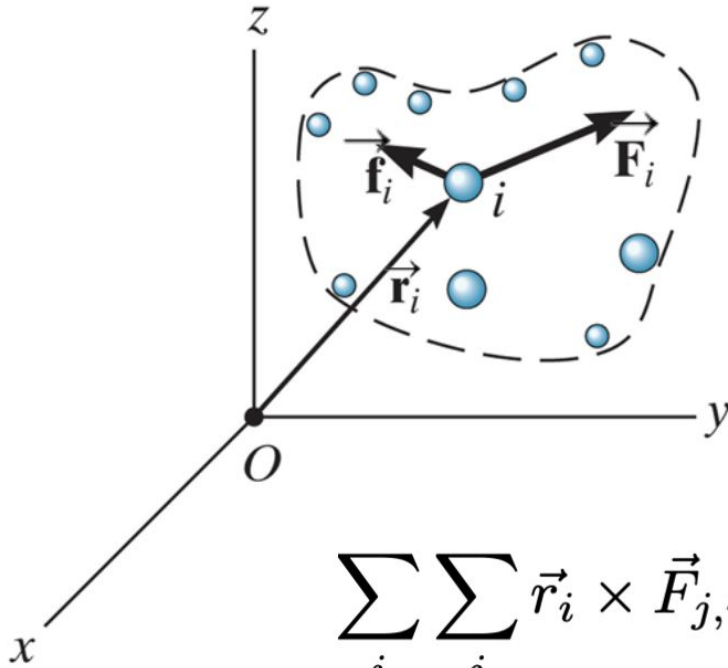
Since $\vec{f}_{1,2} = -\vec{f}_{2,1}$

$$\vec{r}_1 \times \vec{F}_1 + \vec{r}_2 \times \vec{F}_2 + (\vec{r}_1 - \vec{r}_2) \times \vec{f}_{2,1} = \frac{d}{dt} (\vec{L}_{1,o} + \vec{L}_{2,o})$$

If $\vec{r}_{12} \parallel \vec{f}_{2,1}$

$$\vec{r}_1 \times \vec{F}_1 + \vec{r}_2 \times \vec{F}_2 = \frac{d}{dt} (\vec{L}_{1,o} + \vec{L}_{2,o})$$

Angular momentum of a system of particles



The net angular momentum for more particles:

$$\sum_i \sum_j \vec{r}_i \times \vec{F}_{j,i} + \sum_i \sum_{j \neq i} \vec{r}_i \times \vec{f}_{j,i} = \frac{d}{dt} \sum_i \vec{L}_{i,o}$$

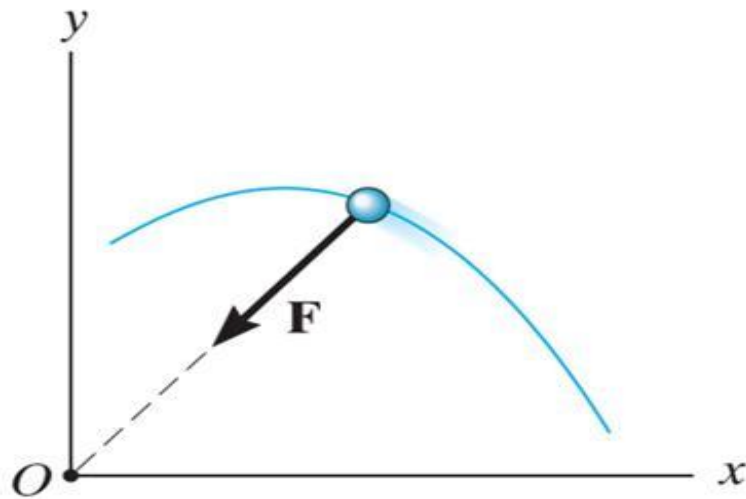
For more particles, the internal net torque (the second term on LHS) is zero because the internal forces occur in equal but opposite *collinear* pairs.

$$\sum_i \vec{\tau}_{i,o} \equiv \sum_i \sum_j \vec{r}_i \times \vec{F}_{j,i} = \frac{d}{dt} \sum_i \vec{L}_{i,o}$$

Conservation of angular momentum

When the sum of angular impulses acting on a particle or a system of particles is zero during the time t_1 to t_2 , the angular momentum is conserved. Thus,

$$(\mathbf{L}_O)_1 = (\mathbf{L}_O)_2$$



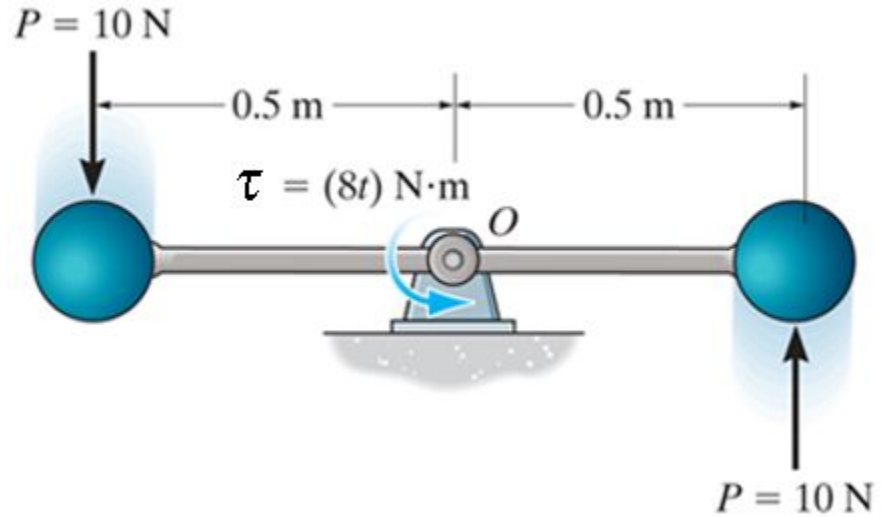
An example of this condition occurs when a particle is subjected only to a central force. In the figure, the force $\mathbf{F}$ is always directed toward point O. Thus, the angular impulse of $\mathbf{F}$ about O is always zero, and angular momentum of the particle about O is conserved.

Quiz

1. Select the correct expression for the angular momentum of a particle about a point.
A) $\mathbf{r} \times \mathbf{v}$ B) $\mathbf{r} \times (m \mathbf{v})$
C) $\mathbf{v} \times \mathbf{r}$ D) $(m \mathbf{v}) \times \mathbf{r}$

2. The sum of the moments of all external forces acting on a particle is equal to
A) angular momentum of the particle.
B) linear momentum of the particle.
C) time rate of change of angular momentum.
D) time rate of change of linear momentum.

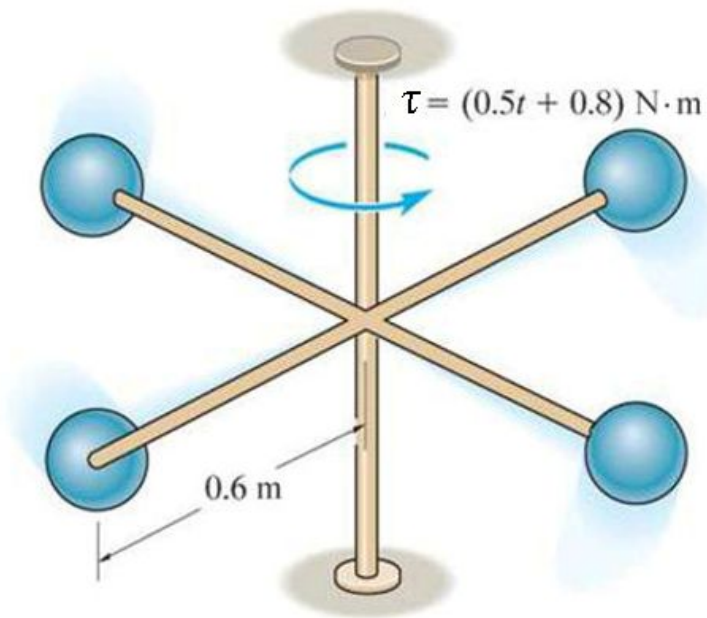
Example



Given: Two identical 10-kg spheres are attached to the rod, which rotates in the horizontal plane. The spheres are subjected to tangential forces of $P = 10\text{ N}$, and the rod is subjected to a couple moment $\tau = (8t)\text{ N}\cdot\text{m}$, where t is in seconds.

Find: The speed of the spheres at $t = 4\text{ s}$, if the system starts from rest.

Example



Given: The four 5-kg spheres are rigidly attached to the crossbar frame, which has a negligible weight.

A moment acts on the shaft as shown, $\tau = 0.5t + 0.8$ ($\text{N}\cdot\text{m}$).

Find: The velocity of the spheres after 4 seconds, starting from rest.

Quiz

1. If a particle moves in the $x - y$ plane, its angular momentum vector is in the
 - A) x direction.
 - B) y direction.
 - C) z direction.
 - D) $x - y$ direction.

2. If there are no external impulses acting on a particle
 - A) only linear momentum is conserved.
 - B) only angular momentum is conserved.
 - C) both linear momentum and angular momentum are conserved.
 - D) neither linear momentum nor angular momentum are conserved.

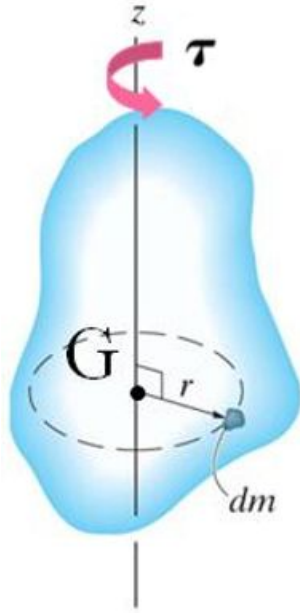
Angular momentum for a rigid body

A rigid body can be viewed as a collection of particles with fixed distances from each other.

Therefore we can apply the above formula to compute its angular momentum directly.

However, it is often useful to introduce the moment of inertia for the discussion of the rigid body angular momentum.

Moment of inertia



Consider a rigid body with a center of mass at G. It is free to rotate about the z axis, which passes through G. Now, if we apply a torque τ about the z axis to the body, the body begins to rotate with an angular acceleration of α .

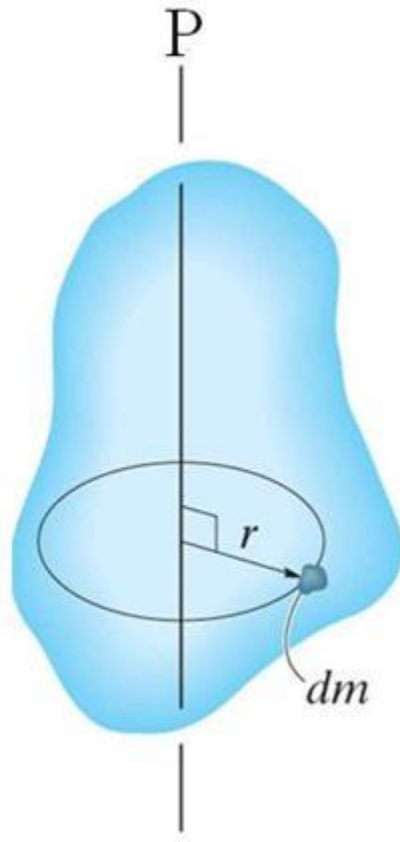
τ and α are related by the equation

$$\tau = I \alpha$$

where I is the mass moment of inertia (MMI) about the z axis.

The MMI of a body is a property that measures the resistance of the body to angular acceleration. The MMI is often used when analyzing rotational motion.

Moment of inertia



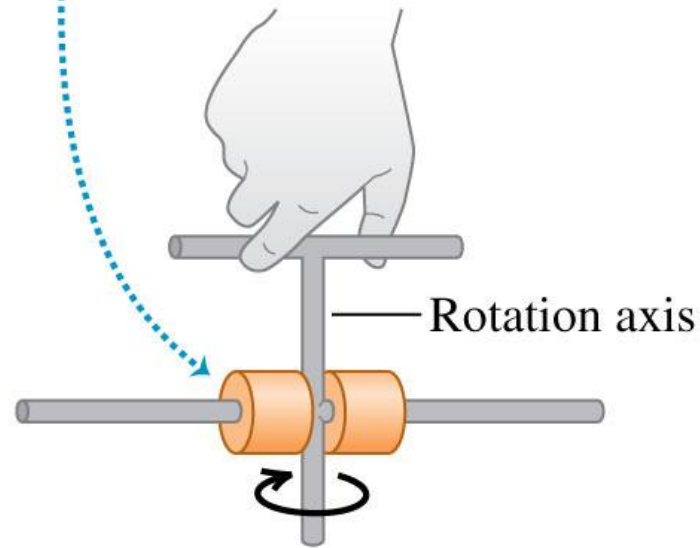
Consider a rigid body and the arbitrary axis P shown in the figure. The MMI about the P axis is defined as $I = \int_m r^2 dm$, where r , the “moment arm,” is the perpendicular distance from the axis to the arbitrary element dm .

The mass moment of inertia is always a positive quantity and has a unit of $\text{kg} \cdot \text{m}^2$.

Moment of inertia

- Here is an apparatus free to rotate around a vertical axis.
- To reduce the moment of inertia, lock the two equal-mass cylinders close to the center of the horizontal shaft.

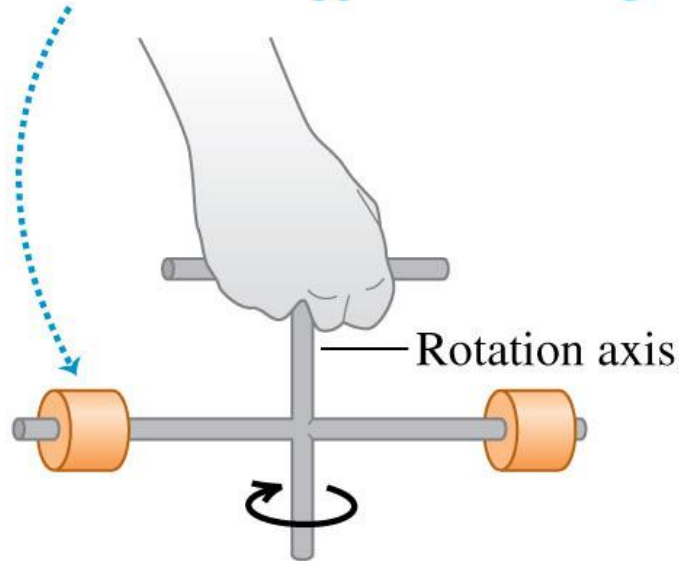
- Mass close to axis
- Small moment of inertia
- Easy to start apparatus rotating



Moment of inertia

- Here is an apparatus free to rotate around a vertical axis.
- To increase the moment of inertia, lock the two equal-mass cylinders far from the center of the horizontal shaft.

- Mass farther from axis
- Greater moment of inertia
- Harder to start apparatus rotating



Moment of inertia of a bird's wing

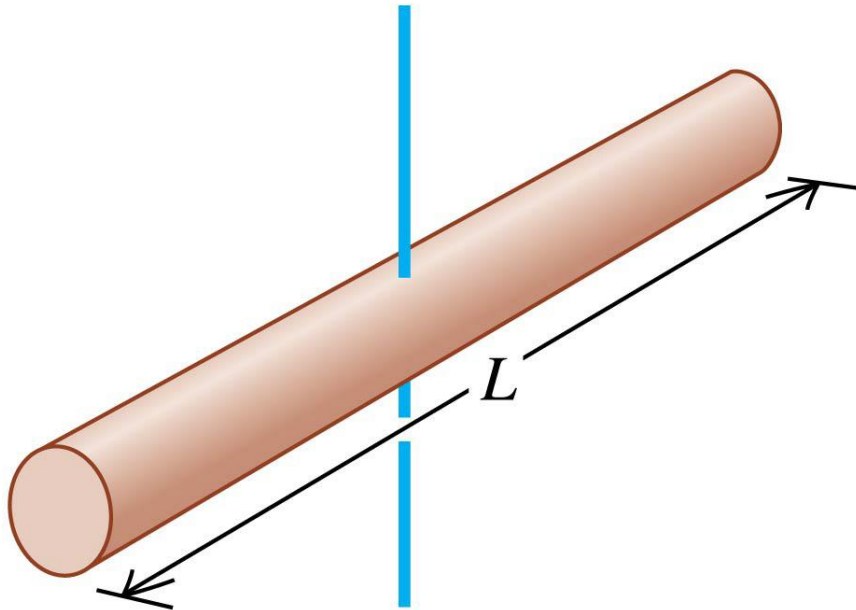
- When a bird flaps its wings, it rotates the wings up and down around the shoulder.
- A hummingbird has small wings with a small moment of inertia, so the bird can move its wings rapidly (up to 70 beats per second).
- By contrast, the Andean condor has immense wings with a large moment of inertia, and flaps its wings at about one beat per second.



Moments of inertia of some common bodies

(a) Slender rod,
axis through center

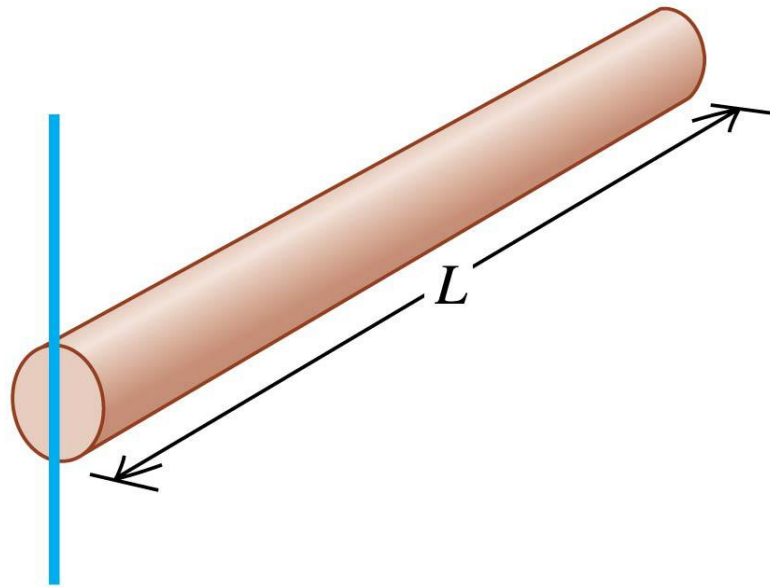
$$I = \frac{1}{12}ML^2$$



Moments of inertia of some common bodies

(b) Slender rod,
axis through one end

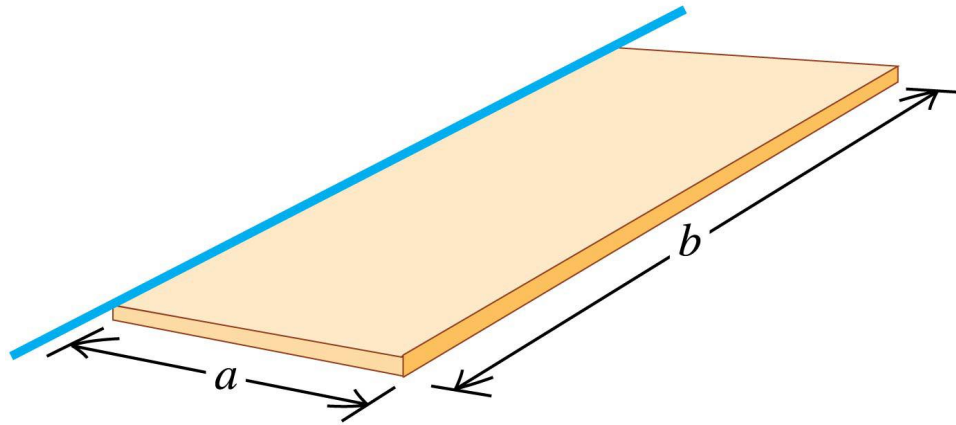
$$I = \frac{1}{3}ML^2$$



Moments of inertia of some common bodies

(d) Thin rectangular plate,
axis along edge

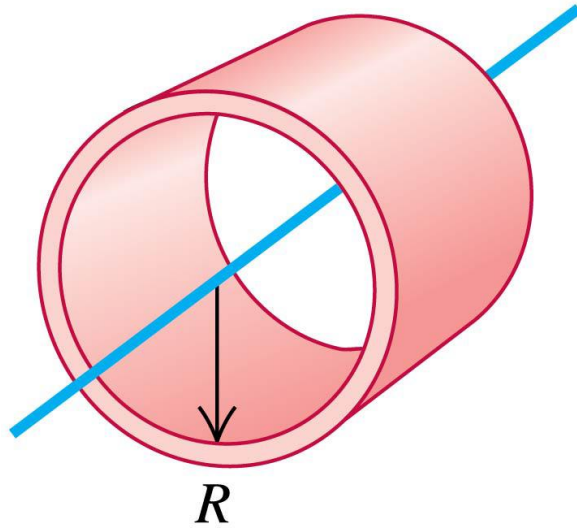
$$I = \frac{1}{3}Ma^2$$



Moments of inertia of some common bodies

(g) Thin-walled hollow cylinder

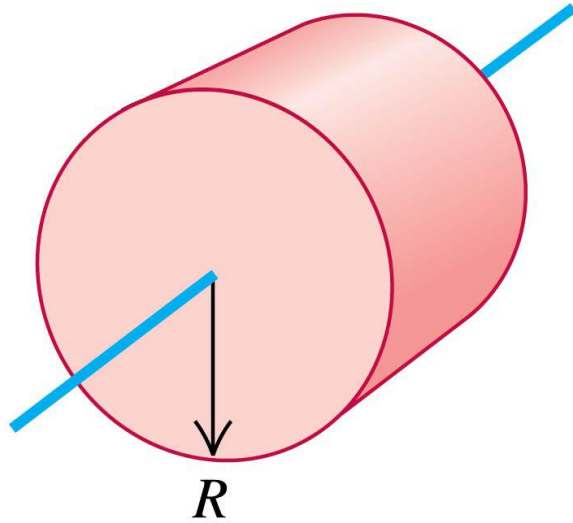
$$I = MR^2$$



Moments of inertia of some common bodies

(f) Solid cylinder

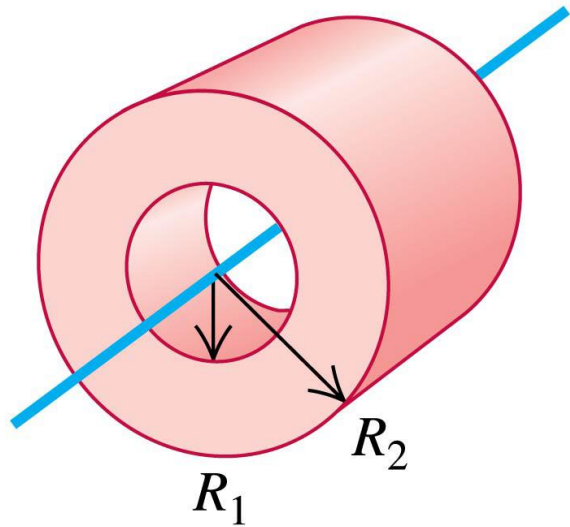
$$I = \frac{1}{2}MR^2$$



Moments of inertia of some common bodies

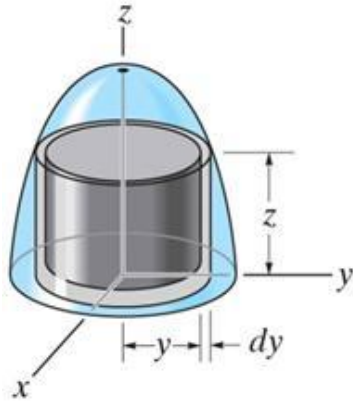
(e) Hollow cylinder

$$I = \frac{1}{2}M(R_1^2 + R_2^2)$$



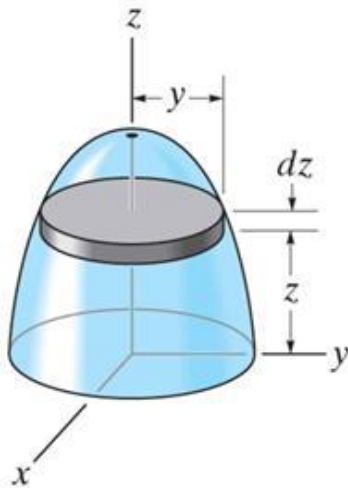
Procedure for analysis

When using direct integration, only symmetric bodies having surfaces generated by revolving a curve about an axis will be considered here.



Shell element

- If a shell element having a height z , radius $r = y$, and thickness dy is chosen for integration, then the volume element is $dV = (2\pi y)(z)dy$.
- This element may be used to find the moment of inertia I_z since the entire element, due to its thinness, lies at the same perpendicular distance y from the z -axis.



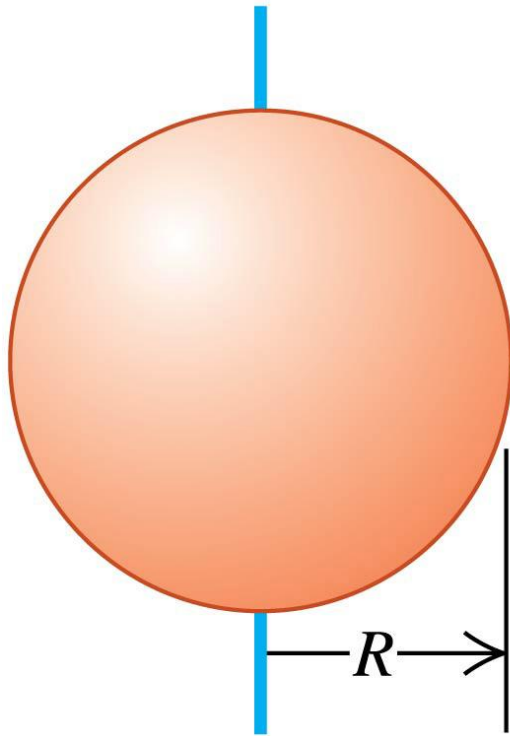
Disk element

- If a disk element having a radius y and a thickness dz is chosen for integration, then the volume $dV = (\pi y^2)dz$.
- Using the moment of inertia of the disk element, we can integrate to determine the moment of inertia of the entire body.

Moments of inertia of some common bodies

(i) Thin-walled hollow sphere

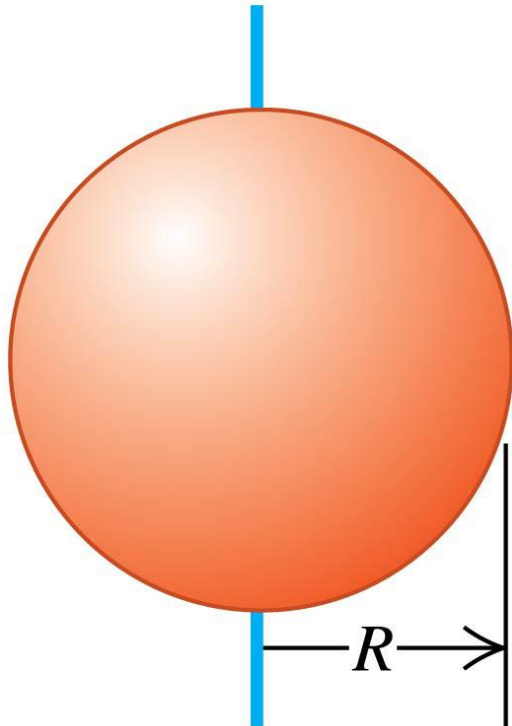
$$I = \frac{2}{3}MR^2$$



Moments of inertia of some common bodies

(h) Solid sphere

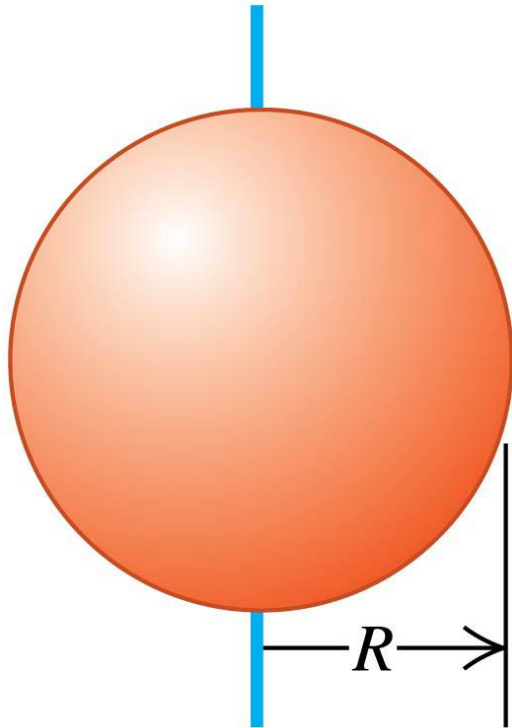
$$I = \frac{2}{5}MR^2$$



Moments of inertia of some common bodies

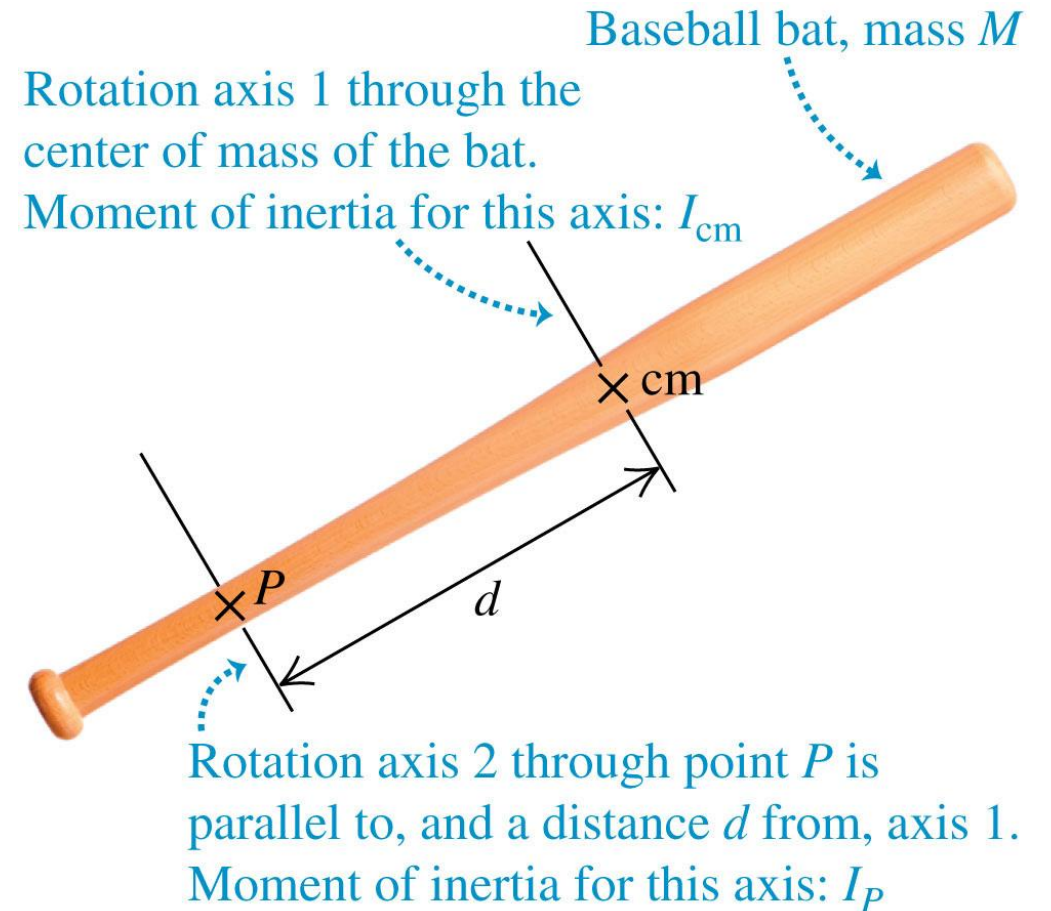
(h) Solid sphere

$$I = \frac{2}{5}MR^2$$



The parallel-axis theorem

- There is a simple relationship, called the **parallel-axis theorem**, between the moment of inertia of a body about an axis through its center of mass and the moment of inertia about any other axis parallel to the original axis.



Parallel-axis theorem: $I_P = I_{\text{cm}} + Md^2$

Parallel-axis theorem

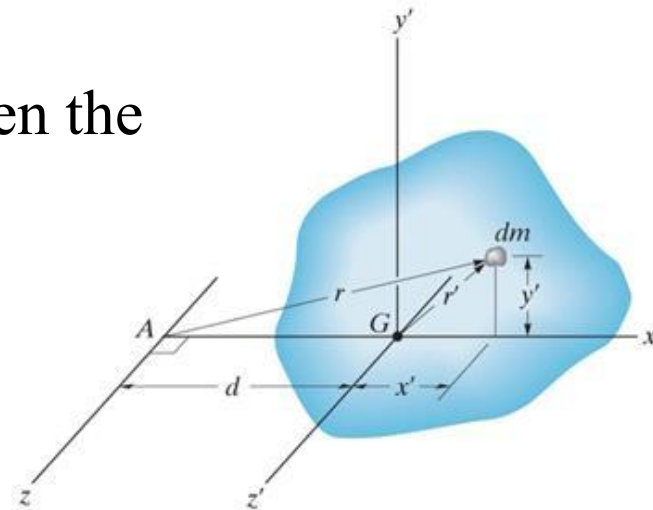
If the mass moment of inertia of a body about an axis passing through the body's mass center is known, then the moment of inertia about any other **parallel axis** may be determined by using the parallel axis theorem,

$$I = I_G + md^2$$

where I_G = mass moment of inertia about the body's mass center

m = mass of the body

d = **perpendicular** distance between the parallel axes



Proof of the parallel-axis theorem

The perpendicular-axis theorem

(This theorem is valid only for pancake objects.)

Consider a pancake object in the x-y plane. Then the perpendicular-axis theorem says that

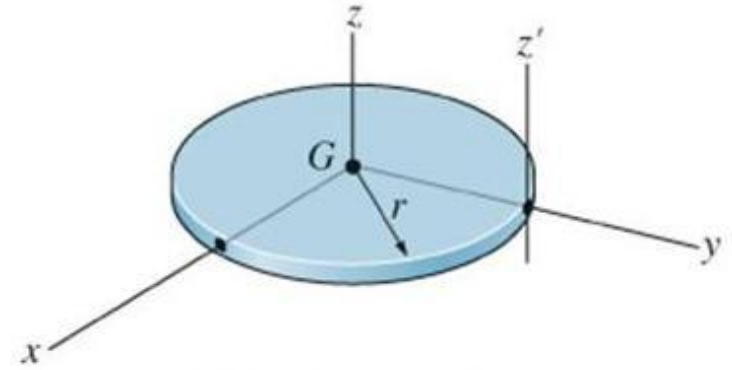
$$I_{zz} = I_{xx} + I_{yy},$$

where

$$I_{zz} = \int (x^2 + y^2) \, dm$$

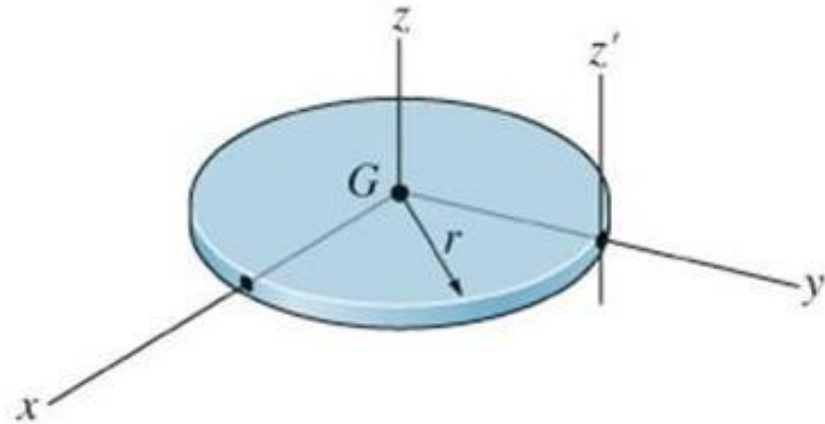
$$I_{xx} = \int (y^2 + z^2) \, dm$$

$$I_{yy} = \int (z^2 + x^2) \, dm$$



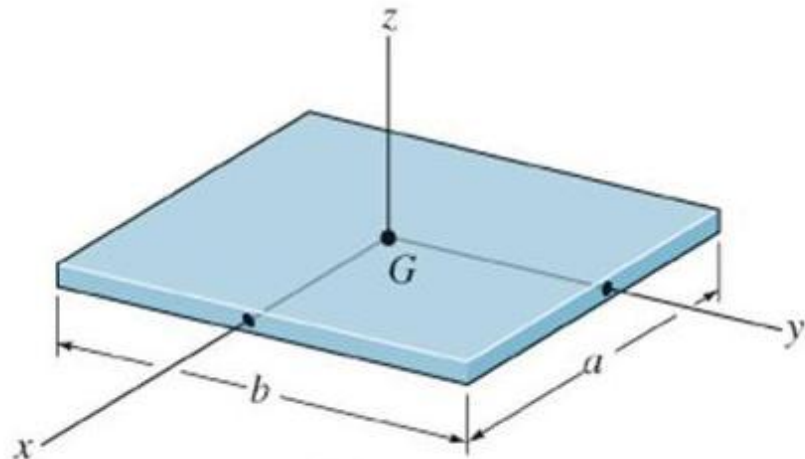
Proof: $z = 0$ for a pancake object.

Moment of inertia



Thin Circular disk

$$I_{xx} = I_{yy} = \frac{1}{4} mr^2 \quad I_{zz} = \frac{1}{2} mr^2 \quad I_{z'z'} = \frac{3}{2} mr^2$$



Thin plate

$$I_{xx} = \frac{1}{12} mb^2 \quad I_{yy} = \frac{1}{12} ma^2 \quad I_{zz} = \frac{1}{12} m(a^2 + b^2)$$

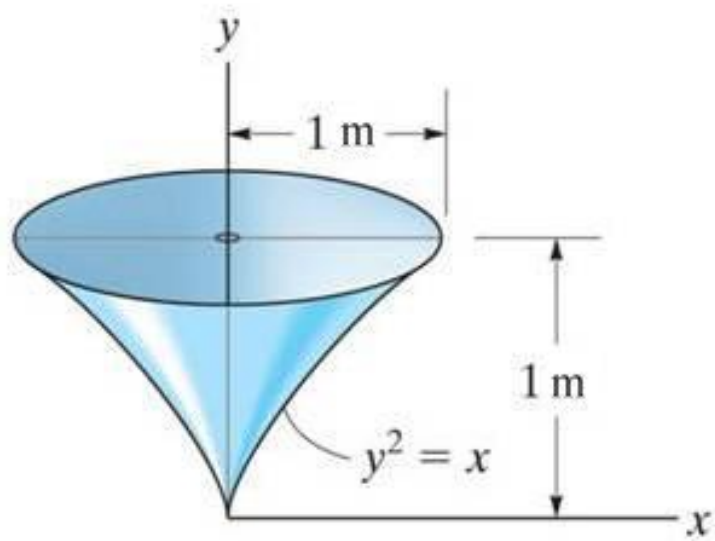
Quiz

1. Mass moment of inertia is a measure of the resistance of a body to _____.
A) translational motion B) deformation
C) angular acceleration D) impulsive motion
2. Mass moment of inertia is **always** _____.
A) a negative quantity
B) a positive quantity
C) an integer value
D) zero about an axis perpendicular to the plane of motion

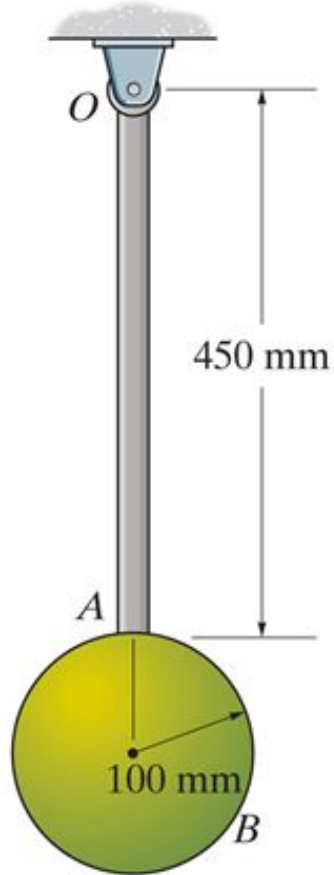
Example

Given: The volume shown with
 $\rho = 3000 \text{ kg/m}^3$.

Find: The mass moment of inertia of
this body about the y-axis.



Example



Given: The pendulum consists of a slender rod with a mass 10 kg and sphere with a mass of 15 kg .

Find: The pendulum's MMI about an axis perpendicular to the screen and passing through point O .

Quiz

1. The mass moment of inertia of a rod of mass m and length L about a transverse axis located at its end is _____.

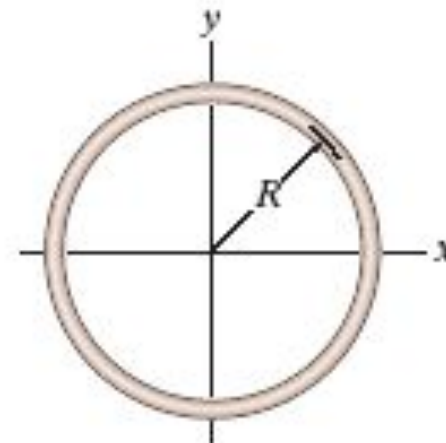
A) $(1/12) m L^2$ B) $(1/6) m L^2$

C) $(1/3) m L^2$ D) $m L^2$

2. The mass moment of inertia of a thin ring of mass m and radius R about the z axis is _____

A) $(1/2) m R^2$ B) $m R^2$

C) $(1/4) m R^2$ D) $2 m R^2$



Quiz

1. The mass moment of inertia of any body about its center of mass is always _____.

- A) maximum B) minimum
- C) zero D) None of the above.

2. If the mass of body A and B are equal but $k_A = 2k_B$, then _____.

- A) $I_A = 2I_B$ B) $I_A = (1/2)I_B$
- C) $I_A = 4I_B$ D) $I_A = (1/4)I_B$

Torque and angular acceleration for a rigid body

- The rotational analog of Newton's second law for a rigid body is:

Rotational analog of Newton's second law for a rigid body:

$$\sum \tau_z = I \alpha_z$$

Net torque on a rigid body about z-axis

Moment of inertia of rigid body about z-axis

Angular acceleration of rigid body about z-axis

- It is easier to loosen or tighten a screw with a larger moment arm.



Combined translation and rotation: dynamics

- The acceleration of the center of mass of a rigid body is:

$$\sum \vec{F}_{\text{ext}} = M \vec{a}_{\text{cm}}$$

Net external force on a body $\rightarrow$ $\sum \vec{F}_{\text{ext}}$ $=$ M (Total mass of body) $\vec{a}_{\text{cm}}$ (Acceleration of center of mass)

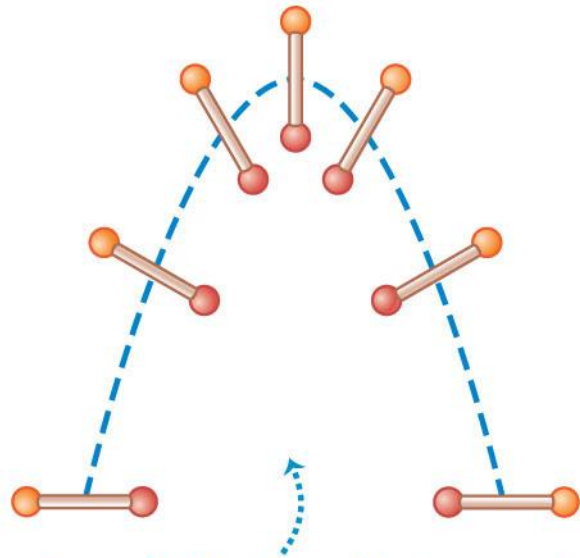
- The rotational motion about the center of mass is described by the rotational analog of Newton's second law:

$$\sum \tau_z = I_{\text{cm}} \alpha_z$$

Net torque on a rigid body about z-axis through center of mass $\rightarrow$ $\sum \tau_z$ $=$ I_{cm} (Moment of inertia of rigid body about z-axis) α_z (Angular acceleration of rigid body about z-axis)

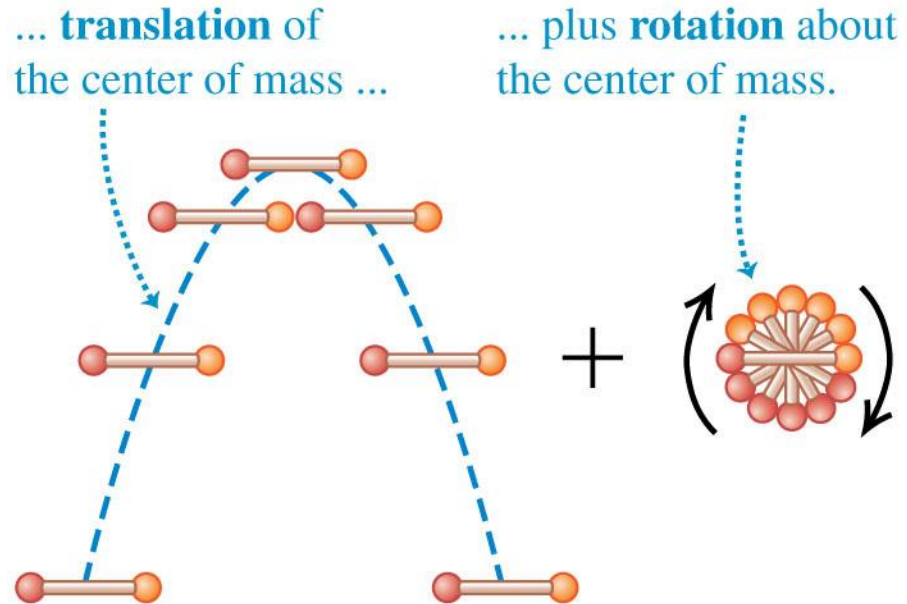
- This is true as long as the axis through the center of mass is an axis of symmetry, and the axis does not change direction.

Rigid body rotation about a moving axis



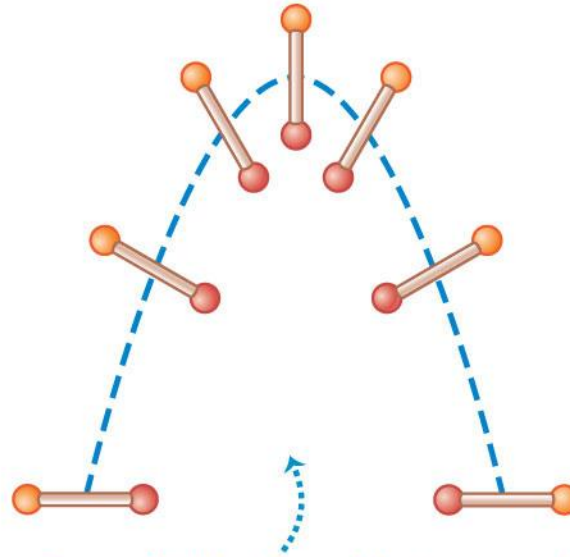
The motion of this tossed baton can be represented as a combination of ...

=



The total energy

- The kinetic energy of a rotating and translating rigid body is $K = \frac{1}{2} M v_{\text{cm}}^2 + \frac{1}{2} I_{\text{cm}} \omega^2$.



The motion of this tossed baton can be represented as a combination of ...

Rotational kinetic energy

- The rotational kinetic energy of a rigid body is:

Rotational kinetic energy of a rigid body rotating around an axis $\rightarrow K = \frac{1}{2}I\omega^2$

Angular speed of body $\rightarrow \omega$

Moment of inertia of body for given rotation axis $\rightarrow I$

- The moment of inertia, I , is obtained by multiplying the mass of each particle by the square of its distance from the axis of rotation and adding these products:

Moment of inertia of a body for a given rotation axis $\rightarrow I = m_1r_1^2 + m_2r_2^2 + \dots = \sum_i m_i r_i^2$

Masses of the particles that make up the body $\rightarrow m_i$

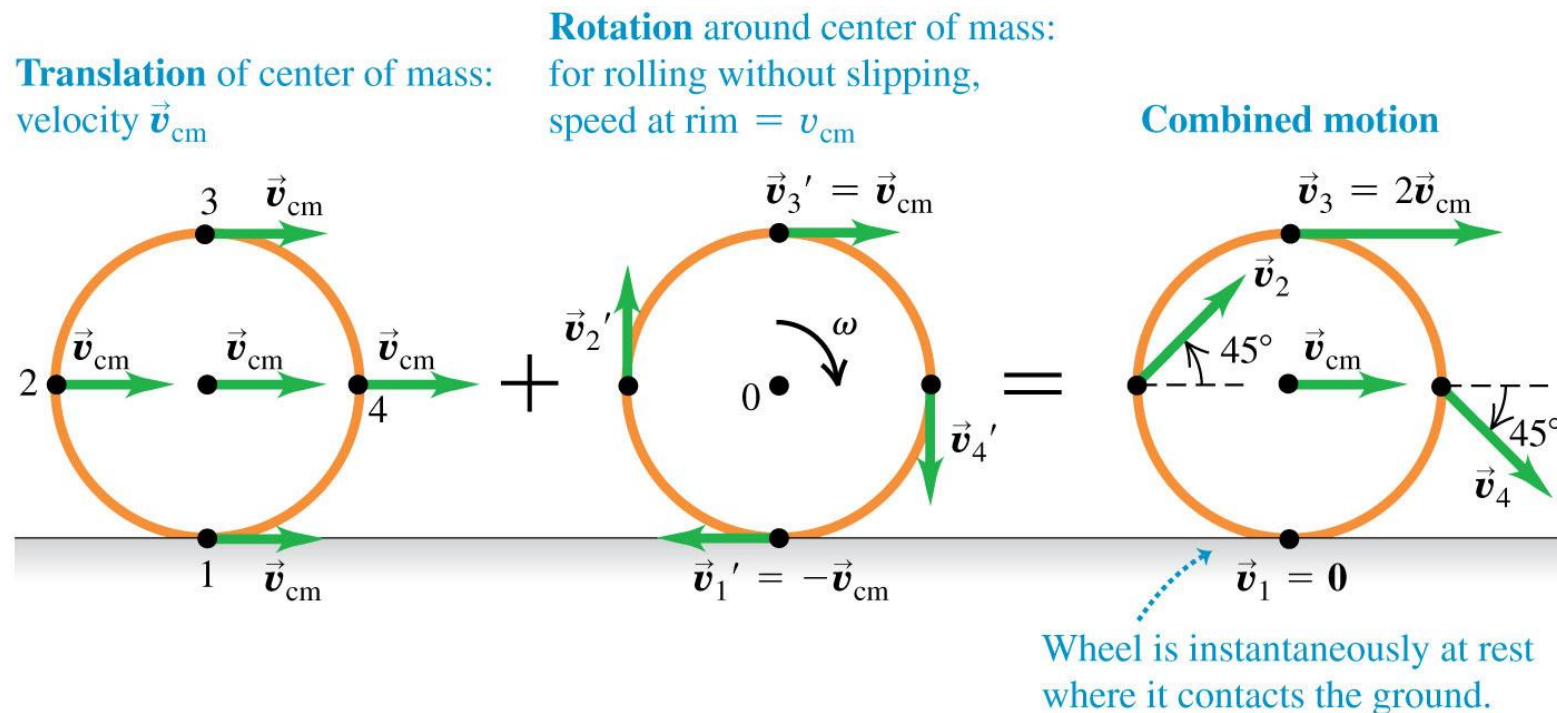
Perpendicular distances of the particles from rotation axis $\rightarrow r_i$

- The SI unit of I is the kilogram-meter² (kg · m²).

Rolling without slipping

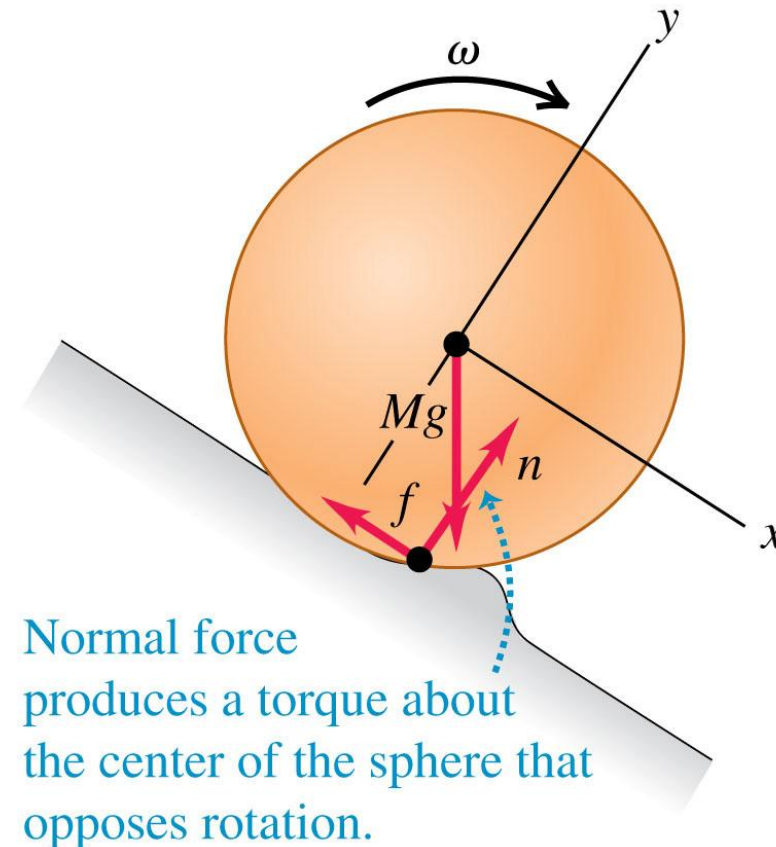
- The motion of a rolling wheel is the sum of the translational motion of the center of mass plus the rotational motion of the wheel around the center of mass.
- The condition for rolling without slipping is

$$v_{\text{cm}} = R\omega.$$



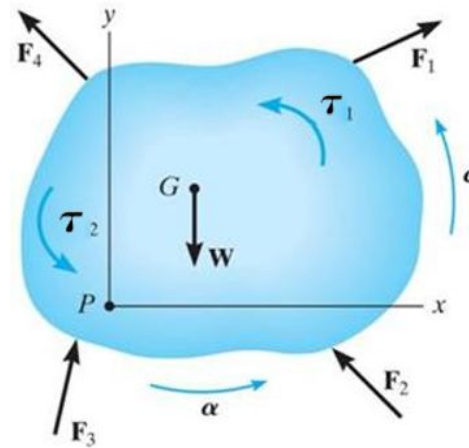
Rolling friction

- We can ignore rolling friction if both the rolling body and the surface over which it rolls are perfectly rigid.
- If the surface or the rolling body deforms, mechanical energy can be lost, slowing the motion.



Planar kinetic equations of motion

- We will limit our study of **planar kinetics** to rigid bodies that are symmetric with respect to a fixed reference plane.
 - As discussed before, when a body is subjected to general plane motion, it undergoes a combination of **translation** and **rotation**.
 - First, a coordinate system with its origin at an arbitrary point P is established.
- The x-y axes should not rotate **but** can either be fixed or translate with constant velocity.

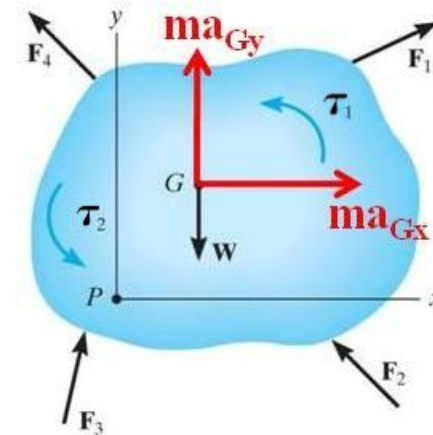


Planar kinetic equations of motion

- If a body undergoes **translational motion**, the **equation of motion** is $\Sigma \mathbf{F} = m \mathbf{a}_G$. This can also be written in scalar form as

$$\Sigma F_x = m(a_G)_x \quad \text{and} \quad \Sigma F_y = m(a_G)_y$$

- In words: the sum of all the external forces acting on the body is equal to the body's mass times the acceleration of its mass center.



Equations of rotational motion

We need to determine the effects caused by the moments of an external force system.

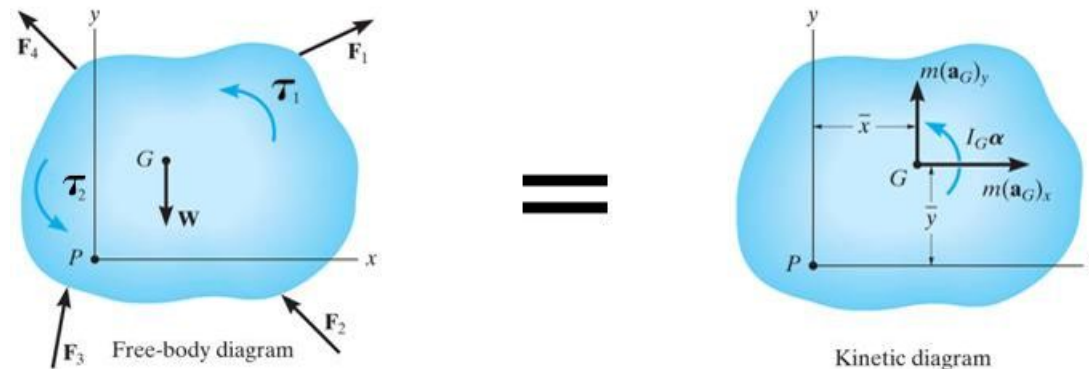
The moment about point P can be written as:

$$\Sigma (\mathbf{r}_i \times \mathbf{F}_i) + \Sigma \boldsymbol{\tau}_i = \langle \mathbf{r} \rangle \times m \mathbf{a}_G + I_G \boldsymbol{\alpha}$$

and $\Sigma \boldsymbol{\tau}_p = \Sigma (\boldsymbol{\tau}_k)_p$

where $\langle \mathbf{r} \rangle = \langle x \rangle \mathbf{i} + \langle y \rangle \mathbf{j}$ and $\Sigma \boldsymbol{\tau}_p$ is the resultant moment about P due to all the external forces.

The term $\Sigma (\boldsymbol{\tau}_k)_p$ is called the **kinetic moment** about point P.



Equations of rotational motion

If point P coincides with the mass center G, this equation reduces to the **scalar equation** of $\Sigma \tau_G = I_G \alpha$.

In words: the resultant (summation) moment about the mass center due to all the external forces is equal to the moment of inertia about G times the angular acceleration of the body.

Thus, **three** independent **scalar** equations of motion may be used to describe the general planar motion of a rigid body. These equations are:

$$\Sigma F_x = m(a_G)_x$$

$$\Sigma F_y = m(a_G)_y$$

$$\text{and } \Sigma \tau_G = I_G \alpha \quad \text{or} \quad \Sigma \tau_p = \Sigma (\tau_k)_p$$

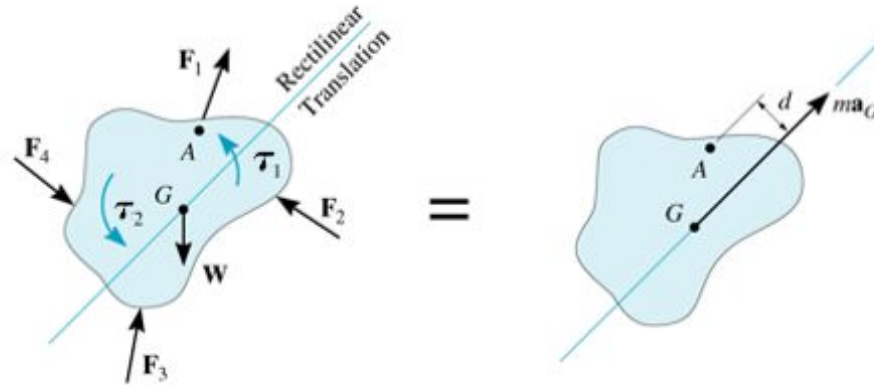
Equations of motion: translation

When a rigid body undergoes **only translation**, all the particles of the body have the same acceleration so $\mathbf{a}_G = \mathbf{a}$ and $\alpha = 0$. The equations of motion become:

$$\Sigma F_x = m(a_G)_x$$

$$\Sigma F_y = m(a_G)_y$$

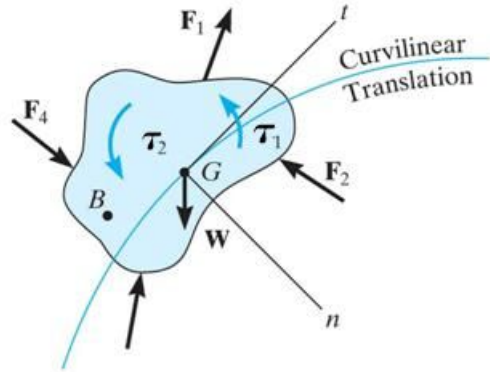
$$\Sigma \tau_G = 0$$



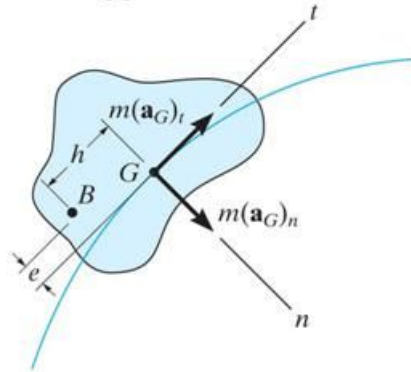
Note that, if it makes the problem easier, the moment equation can be applied about another point instead of the mass center. For example, if point A is chosen,

$$\Sigma \tau_A = (m a_G) d .$$

Equations of motion: curvilinear translation



||



When a rigid body is subjected to **curvilinear translation**, it is best to use an **n-t coordinate system**.

Then apply the equations of motion, as written below, for n-t coordinates.

$$\Sigma F_n = m(a_G)_n$$

$$\Sigma F_t = m(a_G)_t$$

$$\Sigma \tau_G = 0 \quad \text{or}$$

$$\Sigma \tau_B = e[m(a_G)_t] - h[m(a_G)_n]$$

Procedure for analysis

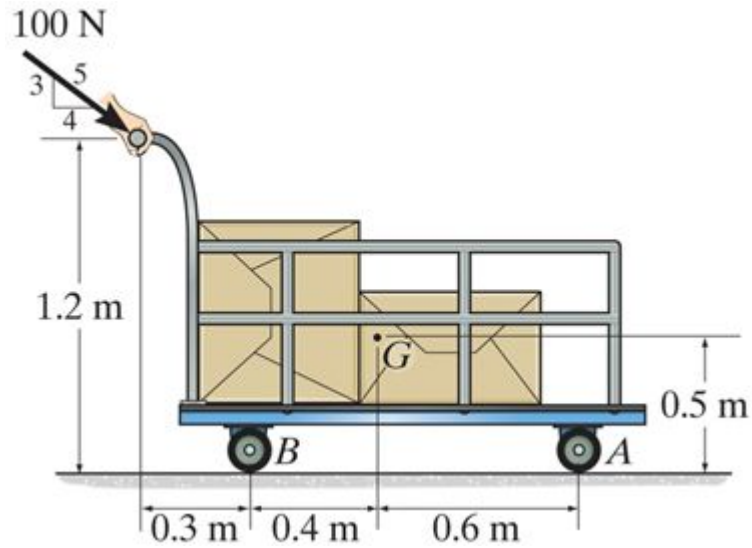
Problems involving kinetics of a rigid body in only translation should be solved using the following procedure:

1. Establish an (x-y) or (n-t) inertial coordinate system and specify the sense and direction of acceleration of the mass center, $\mathbf{a}_G$.
2. Draw a FBD and kinetic diagram showing all external forces, couples and the **inertia forces and couples**.
3. Identify the unknowns.
4. Apply the **three equations of motion** (one set or the other):

$$\left. \begin{array}{l} \Sigma F_x = m(a_G)_x \quad \Sigma F_y = m(a_G)_y \\ \Sigma \tau_G = 0 \quad \text{or} \quad \Sigma \tau_P = \Sigma (\tau_k)_P \end{array} \right| \begin{array}{l} \Sigma F_n = m(a_G)_n \quad \Sigma F_t = m(a_G)_t \\ \Sigma \tau_G = 0 \quad \text{or} \quad \Sigma \tau_P = \Sigma (\tau_k)_P \end{array}$$

5. Remember, friction forces always act on the body **opposing** the motion of the body.

Example



Given: The cart and its load have a total mass of 100 kg and center of mass at G . A force of $P = 100$ N is applied to the handle. Neglect the mass of the wheels.

Find: The normal reactions at each of the two wheels at A and B.

Quiz

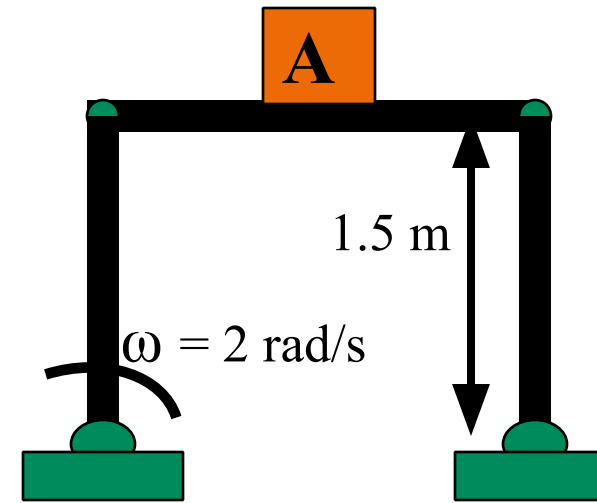
1. As the linkage rotates, box A undergoes _____.

A) general plane motion

B) pure rotation

C) linear translation

D) curvilinear translation



2. The number of independent scalar equations of motion that can be applied to box A is?

A) One B) Two

C) Three D) Four

Work in rotational motion

- A tangential force applied to a rotating body does work on it.

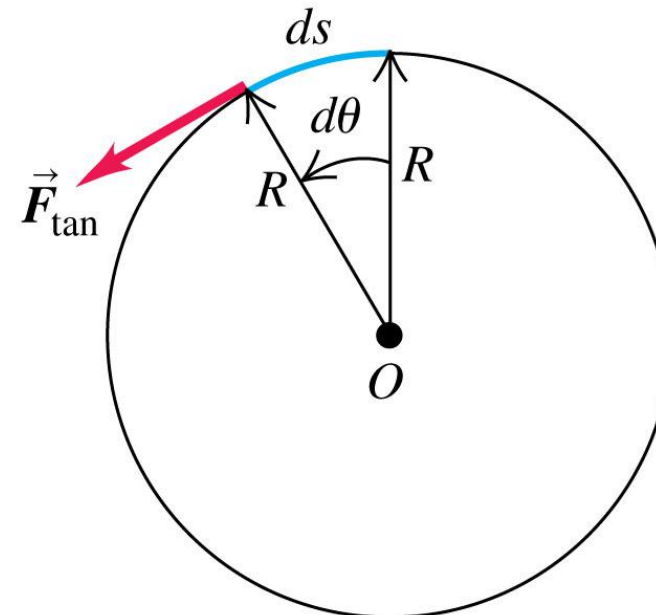
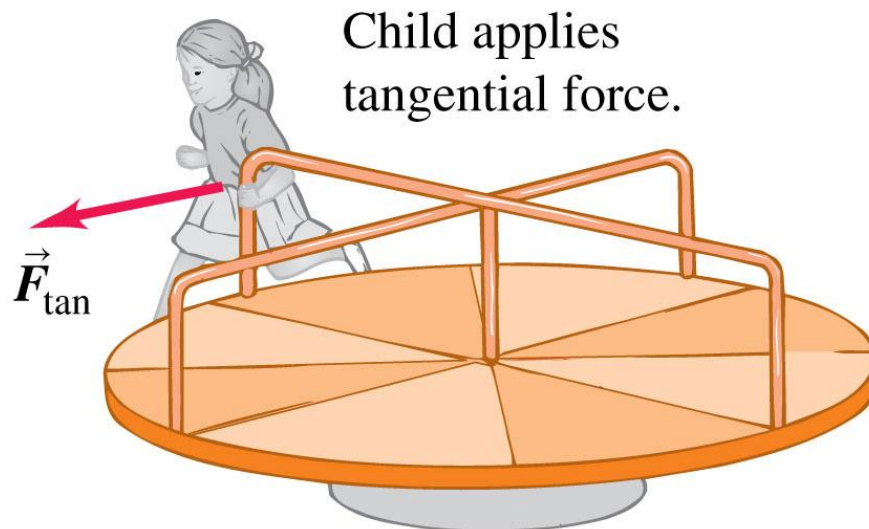
Work done by a torque τ_z

$$W = \int_{\theta_1}^{\theta_2} \tau_z d\theta$$

Upper limit = final angular position
Lower limit = initial angular position
Integral of the torque with respect to angle

(b) Overhead view of merry-go-round

(a)



Work and power in rotational motion

- The total work done on a body by the torque is equal to the change in rotational kinetic energy of the body, and the power due to a torque is:

$$P = \tau_z \omega_z$$

Power due to a torque acting on a rigid body

Torque with respect to body's rotation axis

Angular velocity of body about axis

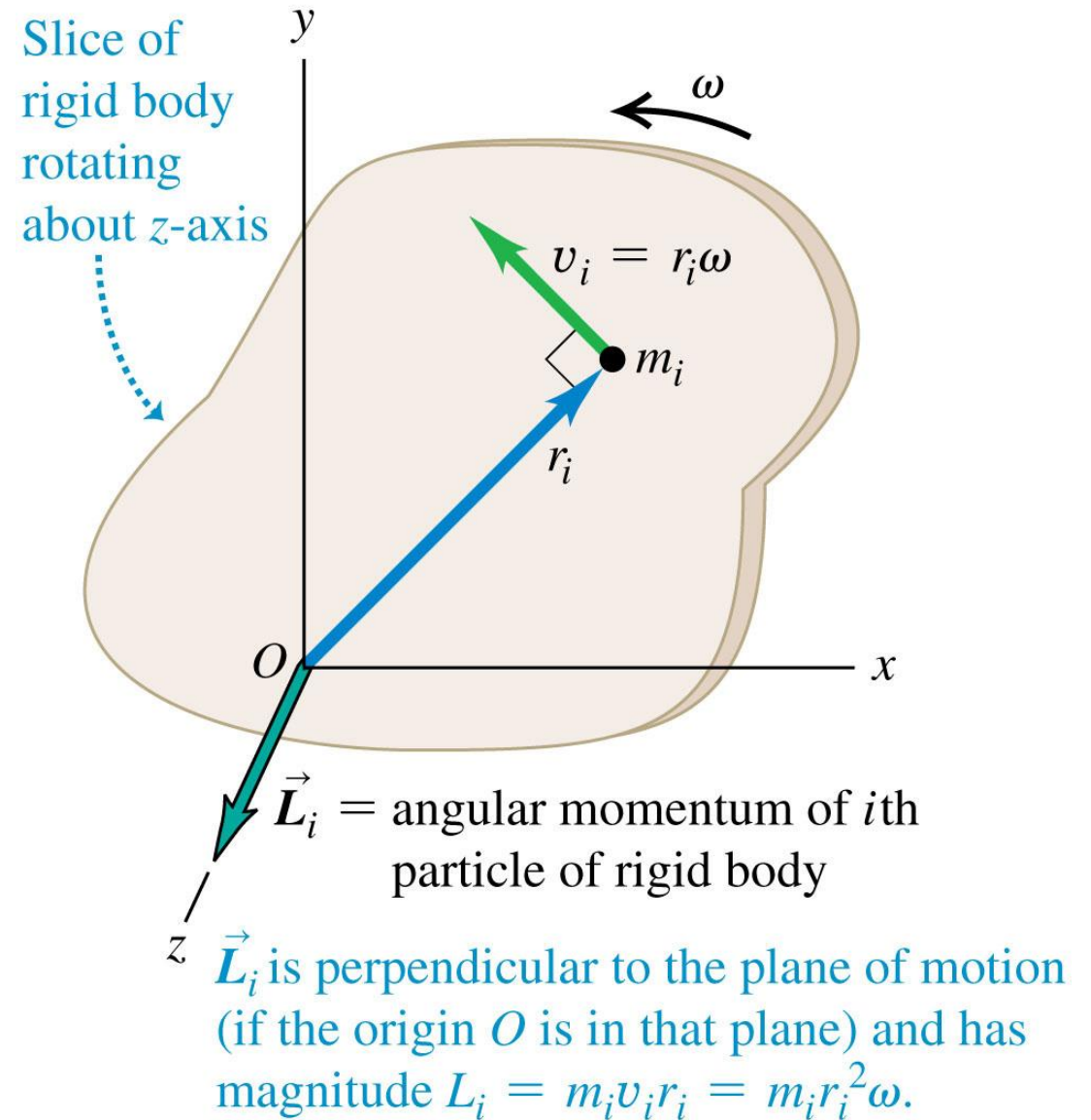
- When a helicopter's main rotor is spinning at a constant rate, positive work is done on the rotor by the engine and negative work is done on it by air resistance.
- Hence the net work being done is zero and the kinetic energy remains constant.



Angular momentum

- To find the total angular momentum of a rigid body rotating with angular speed ω , first consider a thin slice of the body.
- Each particle in the slice with mass m_i has angular momentum:

$$L_i = m_i r_i^2 \omega$$



Angular momentum

- For a rigid body rotating around an axis of symmetry, the angular momentum is:

$$\vec{L} = I\vec{\omega}$$

Angular momentum of a rigid body rotating around a symmetry axis

Moment of inertia of body about symmetry axis

Angular velocity vector of body

- For any system of particles, the rate of change of the total angular momentum equals the sum of the torques of all forces acting on all the particles:

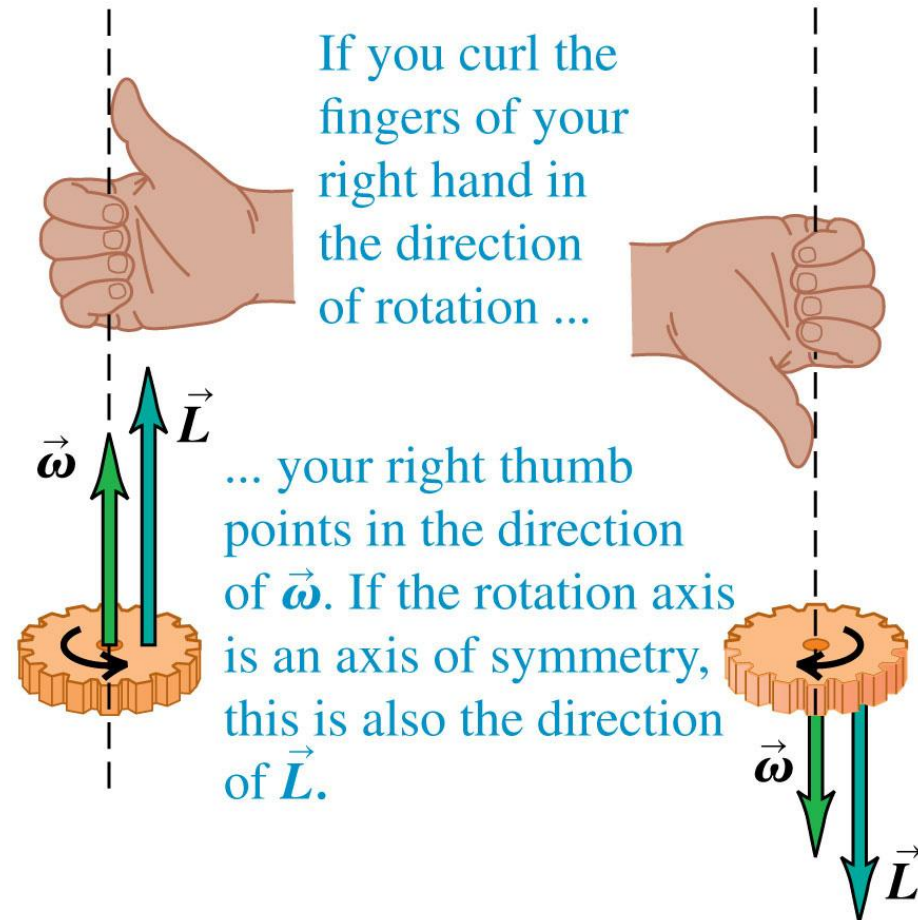
$$\sum \vec{\tau} = \frac{d\vec{L}}{dt}$$

For a system of particles:
Sum of external torques on the system

Rate of change of total angular momentum $\vec{L}$ of system

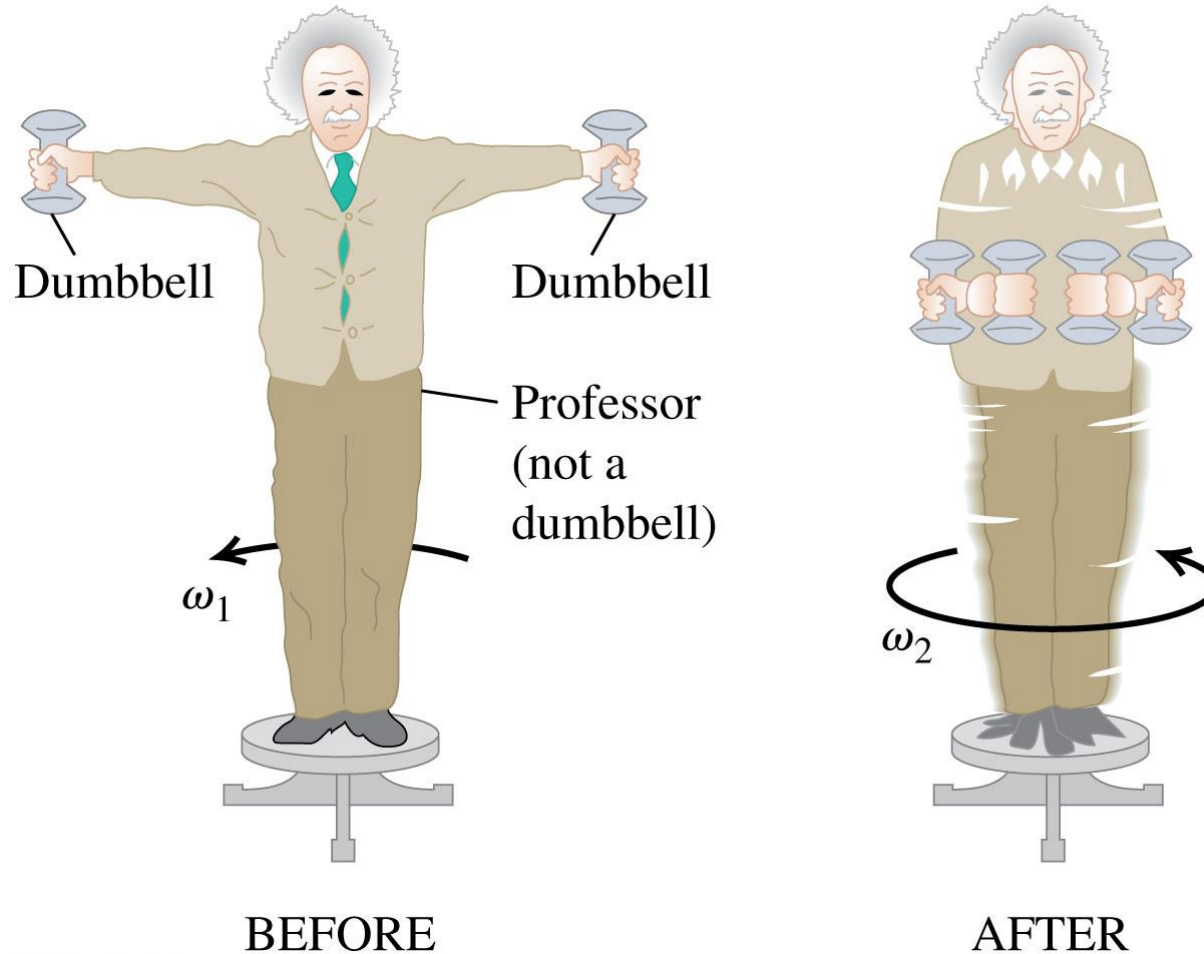
Angular momentum

- The angular momentum of a rigid body rotating about a symmetry axis is parallel to the angular velocity and is given by $\mathbf{L} = I \boldsymbol{\omega}$



Conservation of angular momentum

- When the net external torque acting on a system is zero, the total angular momentum of the system is constant (conserved).



Conservation of angular momentum

- A falling cat twists different parts of its body in different directions so that it lands feet first. At all times during this process the angular momentum of the cat as a whole remains zero.



Summary

Angular momentum

Torque

Principle of angular impulse and momentum

Conservation of angular momentum

Moments of inertia

Angular momentum of a rigid body

Conservation of angular momentum of a rigid body